the energy a particle has can wildly fluctuate so long as this fluctuation is over a short enough time scale. So, just as the accounting system of the airline “allows” you to “borrow” the money for a plane ticket provided you pay it back quickly enough, quantum mechanics allows a particle to “borrow” energy so long as it can relinquish it within a time frame determined by Heisenberg’s uncertainty principle.

The mathematics of quantum mechanics shows that the greater the energy barrier, the lower the probability that this creative microscopic accounting will actually occur. But for microscopic particles facing a concrete slab, they can and sometimes do borrow enough energy to do what is impossible from the standpoint of classical physics—momentarily penetrate and tunnel through a region that they do not initially have enough energy to enter. As the objects we study become increasingly complicated, consisting of more and more particle constituents, such quantum tunneling can still occur, but it becomes very unlikely since all of the individual particles must be lucky enough to tunnel together. But the shocking episodes of George’s disappearing cigar, of an ice cube passing right through the wall of a glass, and of George and Gracie’s passing right through a wall of the bar, can happen. In a fantasy land such as the H-Bar, in which we imagine that $\hbar$ is large, such quantum tunneling is commonplace. But the probability rules of quantum mechanics—and, in particular, the actual smallness of $\hbar$ in the real world—show that if you walked into a solid wall every second, you would have to wait longer than the current age of the universe to have a good chance of passing through it on one of your attempts. With eternal patience (and longevity), though, you could—sooner or later—emerge on the other side.

The uncertainty principle captures the heart of quantum mechanics. Features that we normally think of as being so basic as to be beyond question—that objects have definite positions and speeds and that they have definite energies at definite moments—are now seen as mere artifacts of Planck’s constant being so tiny on the scales of the everyday world. Of prime importance is that when this quantum realization is applied to the fabric of spacetime, it shows fatal imperfections in the “stitches of gravity” and leads us to the third and primary conflict physics has faced during the past century.